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ORIGINAL RESEARCH

Spatiotemporal Crime Prediction and Forecasting Using Machine Learning and Deep Learning Models

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Abstract

Urban crime prediction requires robust, detailed structures capable of capturing both spatial and temporal dynamics while remaining fair and interpreted. This paper introduces a Hierarchical Spatiotemporal Transformer (HSTT) that uses dynamic graph learning to enhance crime risk estimates for practical policing. Geospatial tiling and calendar encoding combined multi-source data such as structured signals, mobility traces, and external factors to generate comprehensive spatiotemporal feature representations. A comparison was made between traditional machine learning methods (like gradient-boosted trees and generalized additive models) and deep learning approaches (including seq2seq transformers and graph neural networks). Results show that HSTT achieved high accuracy with a score of 0.91, an F1-score of 0.90, an AUC of 0.92, and the lowest RMSE at 0.18. Beyond predictive power, HSTT also delivered the best coverage (0.93), the least sharpness (0.15), and the smallest demographic gap (D = 0.04), balancing accuracy and fairness. Although demanding in computational resources, offline training and interpretable results make the framework feasible for real-world application. This study provides an objective perspective on making informed decisions about safer, more transparent city governance by integrating the latest spatiotemporal modeling with ethical considerations.

Keywords Crime prediction · Spatiotemporal forecasting · Machine learning · Deep learning · Hotspot detection

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Abbreviations

s_i	Spatial region (geographical tile) indexed by i
t_j	Discrete time interval indexed by j
$x_{i,j}$	Feature vector of region s_i at time t_j
X	Spatiotemporal feature tensor
d	Dimensionality of feature vectors
$G = (V, E)$	Urban graph with regions as nodes and connections as edges
A, A_{ij}	Adjacency matrix and connectivity strength between regions
D	Degree matrix of the graph
$\widetilde{A}$	Normalized adjacency matrix
$H^{(l)}$	Node embedding at graph layer l
$W^{(l)}$	Learnable weight matrix at layer l
$\sigma(\cdot)$	Activation function (ReLU)
$\phi(\cdot)$	Projection function for dynamic graph learning
Q, K, V	Query, key, and value matrices in attention mechanism
d_k	Dimensionality of attention keys
α_{ij}	Graph attention coefficient between nodes i

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	and j
h_i^T	Temporal embedding of region i
h_i^S	Spatial embedding of region i
α	Fusion weight between temporal and spatial embeddings
$\widehat{y}_{i,j}$	Predicted crime risk for region s_i at time t_j
$y_{i,j}$	Ground-truth crime risk label
L_i, U_i	Lower and upper confidence interval bounds
D	Demographic parity difference (fairness metric)
N, T	Number of spatial regions and time steps

Introduction

City security has become a significant issue for urban centres worldwide, as these centres face not only a loss of trust among the public but also disruptions to their economic and social stability. The conventional policing systems (which are mostly reactionary and resource intensive) are being slowly replaced by predictive systems that allow proactive crime prevention. Predictive policing, which uses historical crime data and signals within an urban environment, has become a feasible solution to allocate resources in a more efficient manner and predict hotspots before their occurrence. However, such a traditional approach to statistics does not consistently capture the spatiotemporal dynamics of urban crime that is preconditioned by dynamically changing demographic, social, and environmental conditions [1]. Recent progress in machine learning (ML) and deep learning (DL) promises new possibilities to create models of crime as a dynamic spatiotemporal process, but not as a constant phenomenon. Initial studies used regression and clustering techniques to determine the crime-prone regions [2], although the current literature focuses on finer-scale predictions in space and time with more sophisticated architectures. It is confirmed that the hegemonic effect on crime patterns is caused by exogenous factors such as weather and other events organised by authorities, socioeconomics, and mobility records [3]. Predictive systems can go beyond reactive forecasting by combining these multi-source datasets into a holistic approach to urban safety.

Spatiotemporal deep learning models, such as Temporal Convolutional Networks (TCNs), recurrent networks, and sequence-to-sequence transformers, have been shown to be outperformed by all other emerging methods regarding long-range dependencies and irregular event dynamics [4]. Graph Neural Networks (GNNs) have been increasingly popular in recent times due to their concept of framing urban space as a mutually dependent structure, with crime at one point potentially leading to risk in neighbouring regions [5]. Such models are useful in the study of localised contagion and cross-region coupling, both of which are core to the propagation of urban crime. In spite of the technical advances, crime forecasting is associated with issues of bias, impartiality and flexibility. There is bias in historical crime information about police, and it is believed that predictive systems will be a silent acceptance of discriminatory measures [6]. Increasing fairness must be systematic and requires such techniques as group fairness audits and differentially private training, which is more focused on accuracy than ethical responsibility [7].

In addition, a city is a dynamical system, and models should be able to evolve over time through retraining and adaptation to other districts and periods [8]. Interpretability is another important dimension. Although deep models offer high predictive accuracy, law enforcement agencies and policymakers need access to how risk estimates were made. Distribution-free confidence interval and feature attribution can be disseminated by providing actionable information, and it will not disrupt the decision-making process [9]. A predictive accuracy framework that is also interpretable is therefore promising to be practical to deploy. The ability to efficiently predict the risk of crime and do so in a responsible manner will also remain a key aspect of the creation of safer and stronger living environments in urban environments as cities grow and digital infrastructure evolves [10]. Although the combination of dynamic deep learning and static spatial features or graph-based learning with fixed temporal windows is common in previous hybrid crime forecasting models, this study presents entirely new dimensions of the model and evaluation. The major new additions are as follows.

(i) The proposed HSTT can learn inter-regional relationships dynamically using changing crime patterns and feature similarity, in contrast to the current hybrid models, which are based on fixed or static spatial graphs, and allow adaptive modeling of non-local crime contagion.

(ii) The framework is unique in that it combines temporal self-attention and graph attention in a hierarchical fusion mechanism, which allows temporary bursts of events, long-term temporal dynamics, and cross-district spatial interaction to be captured simultaneously, a feature that traditional CNN-RNN or Transformer-GNN hybrids do not have.

(iii) Unlike most previous studies that provide point predictions, the proposed model is based on distribution-free confidence intervals that enhance the reliability of decisions made and the operational feasibility of the model in real-world policing.

(iv) This is the unique study to compare the three measures of demographic parity, probability of coverage, and sharpness vs. accuracy indicators, showing that the better the predictive performance, the less bias it has.

In practice it gives deployment rules, informational resource ablation research, prejudice reduction policies and responsive moral practices. In the paradigm of responsible AI, a moderate solution is offered to the paradox between the advancement of technology and the social responsibility of man when predatory policing is embedded within a sensible AI paradigm.

Related and Background Work

In general, Decision trees, RF, SVM, and naive Bayes are all examples of machine learning algorithms used with crime prediction tasks. They can manipulate well-structured bits of data and can offer results in an interpretable format but can be bad at dealing with very non-linear relationships and time constraints. To overcome these constraints, CNNs, RNNs, and LSTM networks are proposed as deep learning architectures to classify and predict crimes. These models are well suited to extract both geographical patterns and temporal trends, as they can automatically extract both spatial and sequential features.

In [11], the authors have shown a high-level crime forecasting model to combine machine learning with prediction and classification-based approaches to improve law enforcement decision-making. The algorithms used in the study include DT, RF, SVM, and neural networks to study past crime trends and forecast future crime patterns. The authors emphasize that it is necessary to use classification in conjunction with forecasting models to not only determine the types of crime but also predict their probability in various places and at various times. They can be more effective in identifying potential areas of high risk and in classifying crime than the traditional statistical methods. Moreover, the study shows that preprocessing methods, such as feature engineering and normalization, play an important role in improving the performance of a model. The products of such a system can guide authorities to conduct proactive policing, efficient resource distribution, and crime prevention approaches.

In [12], the authors investigated the feasibility of developing machine learning algorithms for crime prediction and explored methods to enhance these algorithms for improved safety and proactive law enforcement decision-making. Their study performed statistical analysis of historical crime data to predict crime using decision trees, random forests, naive Bayes, and support vector (SVM) algorithms. The authors observe that proper data preprocessing, including but not limited to cleaning, feature selection and normalization, will be required to enhance accuracy and efficiency. Ensemble techniques, such as random forest, were the best-performing tested models, as they combined a number of decision trees, minimized overfitting, and worked well in complex and noisy data.

In [13], the authors conducted a large review of the existing literature on machine learning applications in crime prediction and highlighted its increasing importance in police practice and community security. The authors examine a broad selection of machine learning methods, including supervised and unsupervised machine learning models, used on crime data to predict criminal activity, forecast hotspots, and predict the behavior of criminals. The research paper notes that DTs, SVM, RF and neural networks are common algorithms used because of their predictive power and because they can be adapted to different datasets. Other challenges the authors refer to are balance of data, bias, privacy issues, and the ethics of predictive policing. Further on, the paper also addresses data preprocessing and feature engineering as approaches to model optimization.

In [14], the authors aimed at predicting the level of cybercrime by applying deep learning algorithms and artificial intelligence algorithms (AI). Due to the rapid development of digital technologies and the Internet, such cybercrimes as phishing, hacking, identity theft, and online fraud have increased significantly and need intelligent detection and prevention mechanisms. The authors suggest a deep learning-based model, processing large volumes of cybercrime data, detecting latent patterns, and forecasting the possibility of a cybercrime incident occurring in the future. The framework incorporates AI methods of feature extraction and classification, which are more accurate than the traditional machine-based learning mechanisms. They report that deep neural networks are superior to classic algorithms in processing and predicting high-dimensional data.

In [15], the authors explored the effectiveness of deep learning structures in categorizing and predicting crimes in terms of their ability to address large and nonhomogeneous pools of them. The paper compares the models implemented in the study: CNNs, RNNs, and LSTM networks with the traditional machine learning methods. The authors emphasize the benefits of deep learning in the automatic extraction of spatial and temporal features with the ability to properly identify the type of crime and predict crime events. There is experimental data demonstrating that LSTM-based designs are most appropriate for processing sequential data, which explains why they are suitable for predicting criminal trends over time. Moreover, CNNs were useful in extracting the geographical trends of spatial information.

In [16], the authors explored interpretable machine learning crime prediction models that consider the paramount balance between predictive performance and model visibility. Although more developed models like deep learning give excellent predictive capability, they are usually viewed as a black box, thus restricting their use and application in policing. To eliminate this, the authors advance explicable methods that not only forecast the crime events but also explain the forecasts. They test models based on urban crime data like decision trees, rule-based learners, and explainable ensemble techniques, as well as methods like SHAP (Shapley Additive Explanations) to analyses feature importance. The research paper notes that crime trends are highly affected by socioeconomic, demographic and spatiotemporal factors. Their findings also indicate that interpretable models can be as competitive as alternative models.

Table 1 presents a succinct critical comparative analysis of the current crime prediction methods and highlights their main limitations. Conventional machine learning models are interpretable and do not reflect complex spatiotemporal relationships. Deep learning models with temporal representations can deal with short-term variations but fail to address long-term variations because they process them sequentially. CNN- and graph-based methods are local spatial pattern-capturing methods that are dependent on fixed structures. Transformer-based models enhance the time model without considering spatial interactions. The proposed HSTT is a unique system that combines temporal attention with dynamic graph learning and responds to fairness and uncertainty, eliminating the fundamental gaps in existing research.

Table 1 Critical comparison of prior crime prediction studies

Prior Approach	Key Strength	Main Limitation / Gap	How HSTT Addresses It
Traditional ML Models (DT, RF, GAM, GBT)	Interpretable, low cost	Cannot capture complex spatiotemporal dependencies	Learns nonlinear temporal and spatial patterns via attention and graphs
Temporal DL Models (LSTM, ConvLSTM)	Short-term temporal modeling	Sequential processing; weak long-range dependency learning	Temporal self-attention enables parallel, long-range modeling
CNN-based Spatiotemporal Models	Local spatio-temporal feature extraction	Fixed receptive fields; limited global context	Multi-head attention captures global temporal dynamics
Graph-based Models (GCN, ST-GCN)	Spatial contagion modeling	Static, predefined graphs	Dynamic graph learning adapts spatial relations over time
Transformer-based Temporal Models	Strong temporal representation	Ignore spatial interdependencies	Hierarchical fusion integrates temporal attention with graph learning
Fairness-unaware Hybrids	High predictive accuracy	Bias amplification; no uncertainty awareness	Fairness evaluation and uncertainty-aware predictions

Proposed Methodology

The proposed model provides an HSTT with dynamic graph learning for a fine-grained crime risk prediction. HSTT also incorporates time and space in a single framework rather than the conventional models that only look at time and space either in a sequence or in a correlation. Structured signals, mobility patterns, and exogenous factors serve as input data for geospatial tiling and calendar coding to produce spatiotemporal feature embeddings. A long-term encoder aware of the irregular dynamics of events and a temporal encoder with multiple heads of self-attention capture the long-term dependencies; a graph attention network (GAT) captures localized contagion and inter-regional interactions. These outputs are combined in a hierarchical manner to yield strong representations of every region-time unit. The last prediction head produces crime risk scores whose confidence interval is distribution-free, making the estimates actionable and uncertain.

Data Integration and Representation

The core component of the proposed crime prediction methodology as shown in Fig. 1 involves the integration of diverse data streams into a unified spatiotemporal framework. This consolidation of varied data sources is fundamental to the system's operational design for predicting criminal activity. Geospatial tiling is used to discretize an urban environment into a collection of spatial tiles. $=\{s_1, s_2, \ldots, s_N\}$ }, and calendar coding is used to discretize a set of temporal sequences. $T=\{t_1, t_2, \ldots ., t_M\}$. The spatiotemporal unit (s_i, t_j) is modelled by a feature vector that includes structural, mobility and exogenous statistics. Formally, the feature of the region s_i at time t_j is defined as:

$$x_{i,j} = [f_{i,j}^{struct}, f_{i,j}^{mob}, f_{i,j}^{exo}] \quad (1)$$

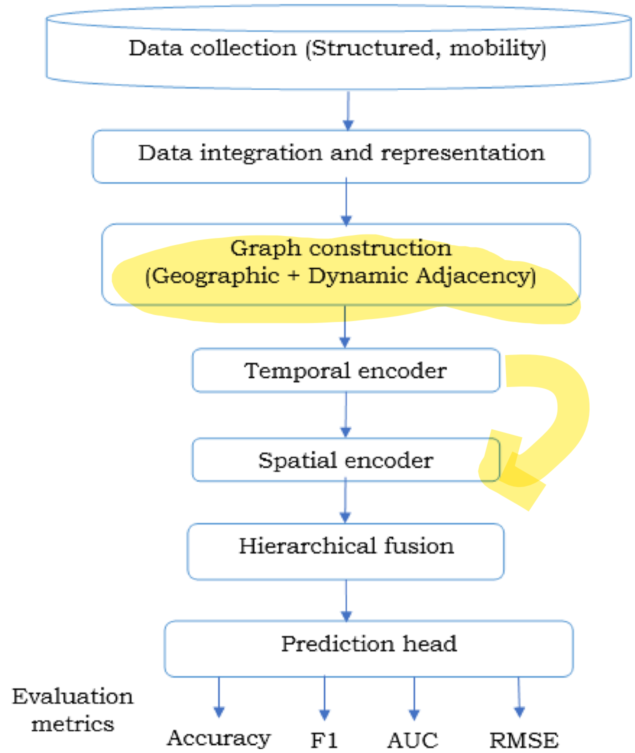

Fig. 1 Proposed Model Methodology

where $f_{i,j}^{struct}$ contains past crime rates, population density, and socioeconomic data. $f_{i,j}^{mob}$contains data about human movement aggregated by GPS, transport or communication signals. $f_{i,j}^{exo}$ contains information about context, including weather, holidays, and other social events.

The data is therefore structured in the form of a tensor $X?\mathbb{R}^{N\times M\times d}$, where d is the dimensionality of the joint feature. In order to model spatial dependencies, we construct an adjacency matrix $A?\mathbb{R}^{N\times N}$, Aij, which is a measure of the geographical or functional connectivity of regions. Similarly, sequence windows also record the temporal dependencies that are characterized as $X_i = [x_{i,t-k}, \ldots ., x_{i.t}]$. This combined representation is necessary to ensure that the model accounts for both local dynamics (crime contagion in neighboring areas) and global interactions (cross-district couplings), giving the input structure a strong basis for the following deep spatiotemporal learning.

The proposed framework uses a hybrid method, as it analyses both conventional machine learning and state-of-the-art spatiotemporal deep learning models to predict crime. In order to create a stable crime risk prediction benchmark, the framework uses baseline prediction models, which are explainable, computationally inexpensive, and broadly used by predictive analytics. Of these, Generalized Additive Models (GAMs) and Gradient-Boosted Trees (GBT) are in the spotlight. GBT is an ensemble method that uses decision trees built sequentially, with each tree trying to reduce the residual errors of the others, in order to minimize a desired loss measure. It also supports non-linear relationships and non-homogenous shapes of features and can be used on structured signals, including counts of crime and demographic attributes. Formally, the model prediction may be written as

$$\widehat{y}\left(x\right)=\sum\nolimits_{m=1}^{M}\gamma_{m}h_{m}\left(x\right) \tag{2}$$

where $h_m\left(x\right)$ denotes the m^{th} decision tree and $\gamma_m$represents the corresponding weight.

In comparison, GAMs are an extension of linear models with smooth non-linear predictor functions, and as such are interpretable and transparent in estimating the impact of individual features on outcomes. A single observed risk at location si and time t_j is expected to follow the model as

$$y_{i,t}=\beta_0+\sum\nolimits_{k=1}^{d}f_k\left(x_{i,t}^{(k)}\right)+? \tag{3}$$

where $f_k\left(.\right)$ is a smooth function on the kth feature and e is error. GBT and GAM are complementary: GBT is more precise when several features interact, and GAM is interpretable and understandable, which are also critical in sensitive domains such as predictive policing. The baseline models are therefore an important point of comparison to measure the value-added of the more sophisticated deep spatiotemporal constructions.

Temporal Modeling

Risk prediction requires capturing the temporal dynamics of crime events because crime processes tend to be sequential and cyclical and are governed by both short-term dynamics and long-term dependencies. To prevent this, the framework uses Temporal Convolutional Networks (TCNs) and Sequence-to-Sequence (Seq2Seq) Transformers, which are better at capturing time-varying sequences than other standard autoregressive models are. One-dimensional dilated convolution allows TCNs to learn long-range dependencies without the vanishing gradient problem that recursive neural networks (RNNs) have. Given an input sequence $[x_{i,t-k}, \ldots ., x_{i.t}]$, the dilated convolution is as defined:

$$F\left(s\right)=\sum\nolimits_{i=0}^{k-1}f\left(i\right).x_{s-d.i} \tag{4}$$

In this context, d represents the dilation factor, $f(i)$ denotes the convolution kernel, and F(s) indicates the output at step s. This allows the network to grow its receptive field exponentially with depth and therefore to be useful on long temporal horizons. By contrast, Seq2Seq Transformers use self-attention to capture dependencies between all-time steps in a single model, without regard to distance. To compute attention for a sequence embedding denoted as $X?\mathbb{R}^{M\times d}$, the calculation of attention is performed as follows:

$$Attention\,(\mathrm{Q},\ \mathrm{K},\ \mathrm{V}) = softmax\left(\frac{QK^T}{\sqrt{d_0}}\right)V \tag{5}$$

with $Q = XW_Q,\ \ K = XW_Q,\ V = XW_V.$

The mechanism allows the dynamically weighted importance of historical time points of the model, such as non-periodic cycles of crime, such as seasonal outbreaks or event-based anomalies. The framework offers a robust model for predicting temporal crime risk within the multi-dimensional structure of a city by integrating the local effectiveness of Temporal Convolutional Networks (TCNs) with the global dependency modelling capabilities of Transformers.

Graph-Based Modeling

Although the temporal models reflect sequential changes, spatial interdependencies between regions in cities also define crime risk. Crime in a particular area tends to spread to adjoining neighborhoods or functionally linked areas, which is known as spatial contagion. In order to model it, the framework uses Graph Neural Networks (GNNs) that describe the city as a graph G (V, E) where the nodes V={s_1,s_2,…,s_N} are spatial regions and the edges E are geographical/functional connections between them. This connectivity is represented as a coded adjacency matrix $A?\mathbb{R}^{N\times N}$, where A_{ij}>0 is a signal of influence between regions s_i and s_j. The neighborhood aggregation in the graph learning layer is defined as updating node features $H(l)$ at layer l:

$$H^{(l+1)} = \sigma\left(\overset{?}{A}\ H^{(l)}W^{(l)}\right) \tag{6}$$

where A∼=D−1represents a normalized adjacency matrix with degree matrix D, $W(l)$ represents a learner weight matrix, and σ(·) denotes a non-linear activation function like ReLU. This formulation enables each region to respond to signals not only based on local features but also based on local neighbors, thus capturing localized diffusion effects. Furthermore, dynamic graph learning is presented to explain non-local interactions, i.e., interconnections between socio-economically comparable but geographically distant regions. In this case, feature similarity is used to update edge weights in an adaptive way:

$$A_{ij}^{(t)} = softmax\left(\frac{\phi\,(h_i^t)^T\phi\,\left(h_j^t\right)}{\sqrt{d}}\right) \tag{7}$$

where ϕ(·) projects node embeddings to a latent space. The model is an effective way to capture urban crime's complex pattern by combining both fixed and dynamic adjacency structures in a way that addresses two features of geographical proximity and functional similarity.

Hierarchical Spatiotemporal Transformer (HSTT)

In order to integrate spatial and temporal dynamics of the crime dynamics, the proposed framework proposes an HSTT on learning dynamic graphs. HSTT, in contrast to isolated temporal or graph-based models, combines both local and global contagions and long-range temporal correlations to produce strong predictions at irregular event regimes. Formally, every region-time unit corresponds to a feature vector $x_{i,t}$expressed in latent space as $h_{ij}^{(t)}$. This model has two complementary phases:

Temporal Attention Layer

For each region si, the temporal embedding sequence $H_i = [h_{i,t-k},\ldots.,h_{i.t}]$, is processed via a transformer encoder. The multi-head self-attention mechanism is expressed as:

$$\left(\mathrm{MH}-\mathrm{Att}\,(H_i) = \overset{M}{\underset{m=1}{\oplus}}\ \mathrm{softmax}\left(\frac{Q_mK_m^\top}{\sqrt{d_k}}\right)V_m\right) \tag{8}$$

Spatial Graph Attention Layer

Simultaneously, spatial dependencies are modeled via a graph attention network (GAT). For each node iii, the representation is updated as:

$$(h_i^{(l+1)} = \sigma\ \left(\sum\ {}_{j\in\mathcal{N}(i)}\alpha\ _{ij}Wh_j^{(l)}\right)) \tag{9}$$

where attention coefficient $\alpha_{ij}$is defined as:

$$\alpha\ _{ij} = \frac{\exp(\mathrm{LeakyReLU}(a^\top\ [Wh_i\|Wh_j]))}{\sum\ _{k\in\ \aleph(i)}\exp(\mathrm{LeakyReLU}(a^\top\ [\|Wh_i\ \ Wh_k]))} \tag{10}$$

ensuring adaptive weighting of neighboring influences.

Hierarchical Fusion

The final representation is obtained by combining temporal and spatial modules:

$$z_{i,t} = \alpha\,.MH - Att\,(H_i) + (1-\alpha\,)\,.GAT\,(h_{i,t}.A) \tag{11}$$

where $\alpha \in [0,1]$ balances temporal and spatial contributions. This hierarchical fusion allows HSTT to generate crime risk scores with uncertainty intervals, enhancing both predictive power and decision confidence. By jointly learning spatial diffusion and temporal evolution, the HSTT captures complex crime propagation patterns more effectively than standalone deep models.

Although the constituent parts of the HSTT (temporal self-attention and graph attention) are motivated by the standard Transformer and GNN structures and are not applied separately or in a sequence, as in previous models, the difference is in the hierarchical coordination and fusion of the parts. In conventional models: Transformers have no spatial connections and work only with temporal sequences. The GNN/ST-GCN is based on the premise of a graph in which the time component is treated shallowly. Existing hybrids usually combine features in both time and space following independent learning, which weakens the interaction between modules, Eq. (11) rewrite as

$$h_i = \alpha\,,\, h_i^{\{T\}} + (1-\alpha)\,,\, h_i^{\{S\}} \tag{12}$$

where $h_i^{\{T\}}$ captures temporal evolution and $h_i^{\{S\}}$ captures spatial contagion.

In contrast, Fig. 2 shows the entire design of the proposed HIST for crime risk prediction. Parallel processing of multi-source input data occurs in the form of temporal, spatial, and contextual features. The temporal self-attention encoder learns long-range and irregular temporal patterns in each region, whereas the dynamic graph attention module learns adaptive spatial relationships and the diffusion of crime across regions. The complementary representations are fused together with a hierarchical fusion layer, which responds to temporal and spatial information to produce a strong crime risk prediction. Finally, the prediction and uncertainty head provides crime risk scores and confidence intervals, which support correct and responsible decision-making.

This fusion is trained throughout enabling the model to focus on the temporal structure in the stable parts and spatial diffusion in the volatile parts, which conventional models cannot do adaptively. A prediction head with uncertainty estimation is then applied to the fused representation, and hierarchical representation learning is directly connected to the enhanced ability to achieve better results in terms of accuracy, calibration, and fairness.

Baseline Models

GBT: These are ensemble learners which progressively train decision trees on the residual, which minimizes a differentiable loss, through gradient boosting; they model intricate non-linear interactions, they deliver feature-importance scores, and they enable relatively rapid training with solid out-of-sample generalization [17].

GAM: These are linear models to which smooth non-parametric functions of predictors are added to be estimated with interpretability and feature-wise, and they are less sensitized to high interactivity tabular effects; they are more transparent and diagnostic and less affected by high interactivity tabular effects [18].

Seq2Seq Transformer: Multi-head self-attention is used in sequence-to-sequence Transformers, which are trained to learn dependencies over entire input and output sequences, allowing them to be trained in parallel and model long-range temporal dynamics; they are more effective than recurrent architectures on tasks that require sequences [19].

GNN: This scale up neural layers to graph-structured data through neighborhood aggregation to support spatial relation learning across regions; they can learn both localized and non-local similarities but are sensitive to graph construction and must be normalized [20].

Fairness Evaluation of HSTT

All sensitive features considered during the fairness assessment process, such as race, age group, and residential

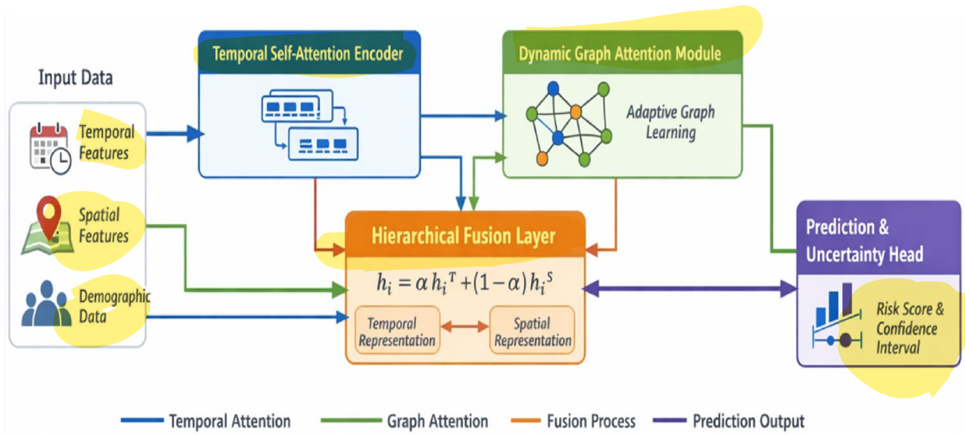

Fig. 2 Hierarchical fusion model overview

location, were anonymized and homogenized before training the models. In particular, the proposed HSTT does not consider raw, sensitive, individual-level identifiers. Rather, demographic data are applied to post hoc fairness at the aggregate spatial or group level. To train the models, the only input features were crime counts, time patterns, mobility statistics, and contextual variables (i.e., weather and calendar effects).

To avoid the model learning discriminatory relationships, the predictive feature set did not include sensitive attributes. Spatial anonymization was achieved by representing location information using coarse-grained geospatial tiles and not specific addresses, which minimized the risk of re-identification. Demographic parity (D) was calculated by comparing the model results among anonymized demographic groups in every spatial unit. This distinction between training characteristics and fairness auditing variables guarantees that the fairness improvements observed when using HSTT are due to the modeling ability of the program and not the direct exploitation of sensitive data. This design is in line with ethical AI principles, as it assists in evaluating bias without encoding protected qualities into the learning algorithm, thus increasing transparency, privacy, and responsible usage.

Ethical Considerations: Privacy, Accountability, and Bias Mitigation

Privacy, accountability, and bias are issues that must be considered when using predictive crime forecasting in law enforcement. Privacy is also safeguarded in this study, where all the data are anonymized and aggregated before modeling, and no personally identifiable information is applied. Spatial data are modeled by coarse-grained geospatial tiles, and sensitive demographic features (e.g., race and age) are not trained and are only used to evaluate post hoc fairness, minimizing the risks of re-identifying and misusing them. To assist model accountability, the proposed HSTT was implemented as a decision support tool, as opposed to an automated enforcement facility. This allows the use of confidence intervals, comparative baselines, and open reporting of performance, which permits informed human control and deters reliance on expectations. Bias reduction is addressed by not considering sensitive variables as inputs into the models and explicitly considering fairness through demographic parity. The hierarchical spatiotemporal design also constrains the amplified bias by acquiring broader patterns instead of local and potentially biased signals. Fairness audits should be conducted regularly, and retraining should be conducted periodically to enable responsible real-world implementation.

Novelty of HSTT Over Existing Spatiotemporal Frameworks

The HSTT is radically different in terms of design and computational resources compared to standard spatiotemporal models such as ST-GCN and ConvLSTM. ST-GCN is based on predetermined spatial graphs and temporal convolutions that restrict its capacity to change with the changing interregional interactions of crimes. ConvLSTM learns local spatiotemporal correlations with convolutional recurrence but suffers from small receptive fields and sequential processing. In contrast, HSTT provides a hierarchical division between temporal and spatial learning, with multi-head temporal self-attention models learning long-range temporal dependencies and irregular temporal dependencies, and dynamic graph attention learning time-varying spatial dependencies depending on feature similarity as opposed to adjacency. The hierarchical fusion mechanism allows for the modeling of local contagion and global cross-district effects, which cannot be jointly valued in the existing frameworks. In contrast to ConvLSTM, which is sequential in nature and has a weakness in parallelism, the transformer-based temporal module of HSTT enables the processing of sequences in parallel, enabling better training on large datasets. In comparison to ST-GCN, the dynamic learning of graphs in HSTT eliminates the need to rely heavily on large dense predefined graphs and instead computes useful and adaptive connections, which makes HSTT better in terms of accuracy-cost trade-offs. Because training is conducted offline, a moderated curriculum increment in complexity is reasonable with respect to a much better predictive response and fairness.

Let N is the count of spatial regions (nodes), T is the count of time steps, E is the count of spatial edges, d represents the dimension of hidden features, and H represents the count of attention heads. Table 2 shows a comparison of the computational complexity of each of the Big-O that obviously puts HSTT in a position to compete with ST-GCN and ConvLSTM.

Here, $E' \ll E$ denotes dynamically learned sparse spatial connections.

ConvLSTM is expensive in terms of computational cost (sequential recurrence) and is scalable only to long temporal horizons. Although ST-GCN is efficient, it is based on graphs that are not dynamic, which results in unnecessary computations on weak or insignificant spatial edges. The proposed HSTT is built upon parallel temporal self-attention, which allows for the effective capture of long-range interactions, whereas dynamic graph learning reduces spatial interactions, imposing spatial complexity on only informative edges. Consequently, HSTT provides improved accuracy–efficiency trade-offs, especially with large urban

Table 2 Computational complexity of Spatiotemporal models

Model	Temporal complexity	Spatial complexity	Overall characteristics
ConvLSTM	$\mathcal{O}\left(T \cdot N \cdot d^2\right)$	Implicit (via convolutions)	Sequential processing, limited parallelism, short-range dependencies
ST-GCN	$\mathcal{O}(T \cdot N \cdot d)$	$\mathcal{O}\left(T \cdot E \cdot d\right)$	Fixed spatial graph, local temporal modeling
Proposed HSTT	$\mathcal{O}\left(T^2 \cdot d/H\right)$	$\mathcal{O}\left(T \cdot E\prime \cdot d\right)$	Parallel temporal attention, adaptive sparse graphs

datasets, and is thus better suited for offline training and real-world applications.

Results and Discussion

The experiment architecture was created using Python 3.9, and the following libraries were used: TensorFlow, PyTorch, and Scikit-learn to build models and Matplotlib to plot the results. All the experiments were performed using a Windows 10 (64-bit) workstation with an Intel Core i7 processor, 16 GB RAM, and an NVIDIA RTX 3080 with 10 GB VRAM to speed up the deep learning training. Whereas the application of baseline models required no parameter development, deep models required hyperparameter optimization. The Seq2Seq Transformer was trained to 128 hidden size, 4 attention heads, and a learning rate of 0.001 with Adam. The depth of neighborhood aggregation of GNN layers was 2 and was activated by ReLU. To prevent overfitting, the proposed HSTT had 3 temporal attention layers and 2 graph layers with a batch size of 64 and dropout of 0.3. After 200 epochs, convergence was achieved, which guarantees predictive accuracy and generalization.

NYPD Dataset Description

The NYPD crime data is a massively produced publicly accessible urban crime dataset published by the New York City Police Department and used in crime research and predictive policing studies. The dataset consists of historical reports of crimes at the incident level that have been compiled across New York City boroughs over the years. Each record is linked to a reported crime occurrence and contains quantified data, such as the type/category of crime, date and time when it occurred, the precinct or patrol area, and the geographic location (latitude and longitude). To make them analytical, raw incident locations are usually summarized into spatial units (e.g., precincts or geospatial grid tiles) and time ranges, such as hourly, daily, or weekly windows. This aggregation allows for the termination of spatial crime distribution and time dynamics, such as periodicity, seasonality, and short-term crime bursts. The data represent a wide variety of types of crimes, and binary or multi-class labelling is possible based on the prediction task (e.g., hotspot vs. non-hotspot areas). Because of the sensitivity of crime data, neither identifiers are provided nor location information is coarse-grained during modeling to protect privacy. The abundance, size, and spatiotemporal complexity of the NYPD data render it ideal for testing more complex models, such as the spatiotemporal transformer and graphical model of urban crime prediction. Table 3 provides dataset attributes and its description.

Table 3 NYPD dataset attributes

Attribute	Type	Description
Incident_ID	Categorical	Unique crime record identifier
Crime_Category	Categorical	Offense classification label
Date_Time	Timestamp	Crime occurrence time
Day_Week	Categorical	Weekly temporal indicator
Month_Season	Categorical	Seasonal temporal indicator
Latitude	Continuous	Spatial coordinate (aggregated)
Longitude	Continuous	Spatial coordinate (aggregated)
Precinct_Borough	Categorical	Administrative spatial unit
Location_Type	Categorical	Environmental context
Crime_Count	Integer	Aggregated incidents per region–time window
Risk_Label	Binary / Continuous	Hotspot indicator or crime risk score

Accuracy measures the proportion of correctly predicted crime risk outcomes over total predictions. It is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

where TP, TN, FP, FN represent true positives, true negatives, false positives, and false negatives. High accuracy reflects balanced performance across safe and risky regions.

The F1-Score balances precision and recall, making it vital when crime risk classes are imbalanced. It is computed as:

$$F1 - measure = \frac{2 \times precision \times recall}{precision + recall} \tag{14}$$

This harmonic mean emphasizes both detection ability (recall) and correctness (precision), ensuring the system neither underestimates nor exaggerates crime hotspots.

AUC evaluates the probability that the model ranks a randomly chosen high-risk area higher than a safe area. Formally:

$$AUC = \int_0^1 TPR\left(FPR^{-1}\right)(x)\,dx \tag{15}$$

where TPR is true positive rate and FPR is false positive rate.

RMSE quantifies the average magnitude of prediction errors between estimated and true crime risks. It is expressed as:

$$RMSE = \sqrt{\frac{1}{n}\sum{}_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2} \tag{16}$$

Lower RMSE means more accurate risk intensity prediction, reducing uncertainty in hotspot mapping.

Demographic parity is a fairness metric that states that sensitive populations have equal prediction rates. Lower D implies less bias, and the various demographic groups are treated similarly when making crime risk predictions. The difference is:

$$\Delta = \left|P(\bar{Y} = 1 \mid A = a) - P(\bar{Y} = 1 \mid A = b)\right| \tag{17}$$

Coverage probability measures the accuracy of the predicted confidence intervals (CIs). $1\{\cdot\}$ is the indicator function. A large coverage means that real risks are not out of the range of expected uncertainty and enhances confidence in estimates. It is defined as:

$$Coverage = \frac{1}{n}\sum{}_{i=1}^{n} 1\{y_i \in CI_i\} \tag{18}$$

Sharpness is a measure of concentration of predictive intervals and a measure of informativeness, and U_i and L_i are the upper and lower limits of the confidence interval. Less sharpness means more accurate and fewer vague predictions, without risk estimates. It is given by:

$$Sharpness = \frac{1}{n}\sum{}_{i=1}^{n}(U_i - L_i) \tag{19}$$

Model Selection and Validation

To trade off the computational efficiency and reproducibility, a manual, empirically informed hyperparameter tuning strategy was used to select and validate a model, as opposed to an exhaustive grid search or Bayesian optimization. This is especially the best method to use for deep spatiotemporal architectures, where exhaustive search is simply too costly.

Default library settings were used for baseline machine learning models, that is, GAM. These models are stable, easy to interpret, and maintain standard settings, providing a fair and transparent performance standard compared to more complex deep learning methods. The hyperparameters of the deep learning baselines (Seq2Seq Transformer and GNN) and the proposed HSTT were gradually changed according to their validation performance and convergence. Learning rates with 0.0005 as the lower bound and 0.005 as the upper bound were tested, and 0.001 was chosen to be optimized stably. The hidden dimensions were evaluated between 64 and 128, and 128 was selected to provide an adequate representational space. The number of attention heads was varied to 2, 4, and 8, and 4 heads performed the best trade-off between accuracy and computational cost. Dropout rates between 0.1 and 0.5 were considered, and 0.3 was good for eliminating overfitting. The graph-based components were also adjusted by changing the number of graph layers to 1, 2, or 3, with 2 layers providing the best spatial generalization. Each deep model was trained for a maximum of 200 epochs, and training was stopped when the validation metrics stopped changing to ensure that the best model was selected with no overfitting. Table 4 shows the hyperparameter optimization strategy and ranges tested is given below,

Table 4 Hyperparameter optimization strategy and tested ranges

Model Category	Hyperparameter	Tested range / values	Selected value
GBT	Tree depth	Default (library setting)	Default
	Learning rate	Default	Default
GAM	Smoothing parameters	Automatic (library estimation)	Automatic
Seq2Seq Transformer	Hidden dimension	{64, 128}	128
	Attention heads	{2, 4, 8}	4
	Learning rate	{0.0005, 0.001, 0.005}	0.001
	Dropout rate	{0.1, 0.3, 0.5}	0.3
GNN	Graph layers	{1, 2, 3}	2
	Activation	ReLU	ReLU
Proposed HSTT	Temporal attention layers	{2, 3, 4}	3
	Graph attention layers	{1, 2, 3}	2
	Hidden dimension	{64, 128}	128
	Attention heads	{2, 4, 8}	4
	Dropout rate	{0.1–0.5}	0.3
	Batch size	{32, 64, 128}	64
	Epochs	Up to 200	200

The performance of the models is compared in Table 5 in terms of accuracy, F1-score, AUC, and RMSE. GBT scored 0.78 in accuracy and 0.75 in F1, and GAM scored 0.72 in F1 and 0.74 in accuracy. The Seq2Seq Transformer was raised to 0.82 accuracy, 0.81 F1 and 0.29 RMSE. GNN fared better with an accuracy of 0.85, an F1 of 0.84 and an RMSE of 0.25. The proposed HSTT achieved the highest accuracy, 0.91; the highest F1-score, 0.92; the highest AUC, 0.92; and the lowest RMSE of 0.18, proving the best predictive power of crime risk.

Figure 3 shows the performance of the various models in relation to accuracy, F1-score and AUC. Gradient-boosted trees (GBT) reached 0.78 accuracy, 0.75 F1 and 0.80 AUC, which is medium effective in structured prediction. Generalized Additive Models (GAM) had lower 0.74, 0.72 F1, and 0.76 AUC, indicating that this model cannot reflect non-linear spatiotemporal behavior. The Seq2Seq Transformer became much more accurate, with 0.82 accuracy, 0.81 F1 and 0.83 AUC, which can be explained by its capacity to exploit temporal dynamics. Graph Neural Networks (GNN) also improved performance (0.85 accuracy, 0.84 F1, 0.86 AUC) by modelling the effects of contagion at the neighborhood level. The proposed HSTT was clearly more accurate than all the baselines, with 0.91 accuracy, a 0.90 F1 score, and 0.92 AUC. This high performance has been achieved because the combination of both dynamic graph learning and temporal attention has enabled the representation of both local and global patterns of crime propagation.

Figure 4 displays the RMSE values of the various models, indicating the levels of prediction error. GAM had a large RMSE of 0.37, and GBT had a smaller estimate of crime risk with a strong RMSE of 0.34. The error rate of the Seq2Seq Transformer is reduced to 0.29 by its ability to capture time relationships, and it was even further reduced to 0.25 by GNN, which represents the benefit of learning a spatial neighborhood. The lowest RMSE of 0.18 was obtained with the proposed HSTT, which proves its better accuracy. This drastic reduction can be attributed to its time-space hierarchy that increases the accuracy of fine-grained prediction in random metropolitan crime distributions.

Table 6 gives computational efficiency. GBT took 45s to train and 40s to test, and GAM took the shortest time of 38s and 33s, respectively. Seq2Seq Transformer was more expensive (120s training, 105s testing), and GNN was even more expensive (150s, 125s). The complexity of the proposed HSTT was 210 s of training and 190 s of testing. This trade-off between HSTT and real-world application is compensated by superior predictive accuracy, even though HSTT has a higher runtime.

Figure 5 compares the training and testing times of all models, illustrating trade-offs in computational efficiency. Generalized Additive Models (GAM) were the quickest models and took 38 s to train and 33 s to test due to their simple linear and smooth form of functions. GBTs (Gradient-Boosted Trees) with a training time of 45 s and a testing time of 40 s were added, which has the benefit of learning the ensemble trees. The larger resources needed by the Seq2Seq Transformer (train 120s, test 105s) are corresponding to its sequence-to-sequence history complexity. Neighborhood aggregation and graph construction raised computation to 150s training and 125s testing with Graph Neural Networks (GNNs). The most expensive proposed HSTT cost – 210s

Table 5 Comparison of model performance

Model	Accuracy	F1-Score	AUC	RMSE
GBT	0.78	0.75	0.8	0.34
GAM	0.74	0.72	0.76	0.37
Seq2Seq Transformer	0.82	0.81	0.83	0.29
GNN	0.85	0.84	0.86	0.25
Proposed HSTT	0.91	0.9	0.92	0.18

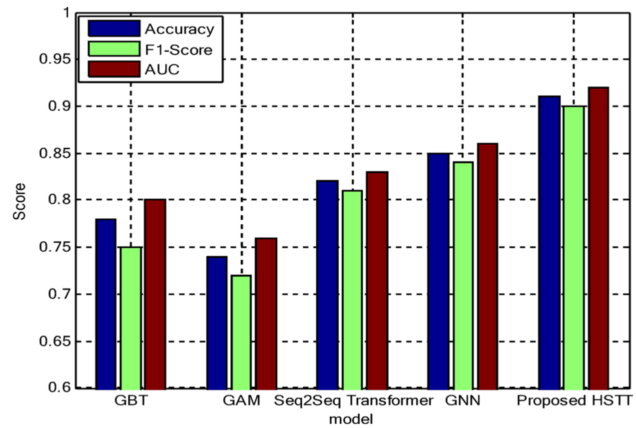

Fig. 3 Comparative Performance of Different Models on the NYPD Dataset

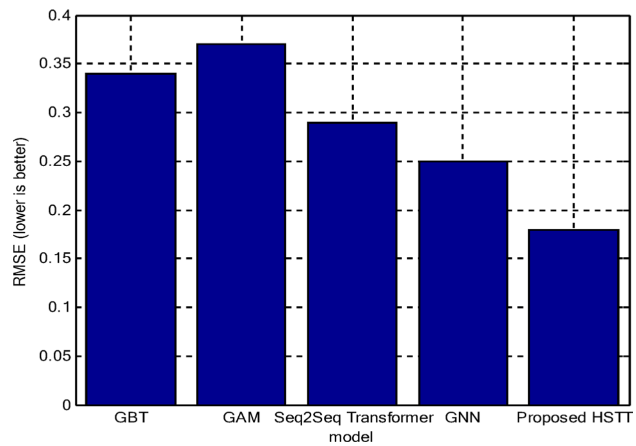

Fig. 4 RMSE Comparison of Crime Risk Prediction Models

Table 6 Training and testing time

Model	Training time (s)	Testing time (s)
GBT	45	40
GAM	38	33
Seq2Seq Transformer	120	105
GNN	150	125
Proposed HSTT	210	190

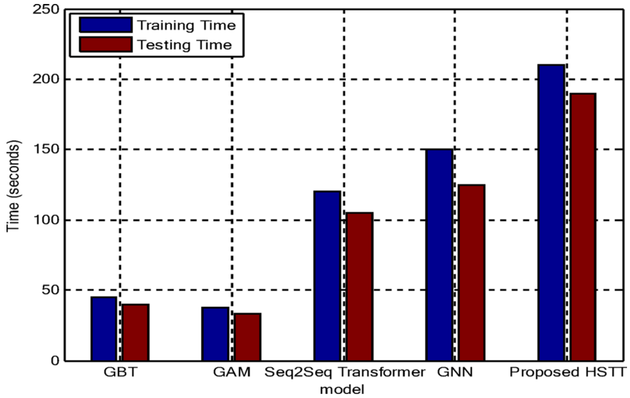

Fig. 5 Training and Testing Time Comparison

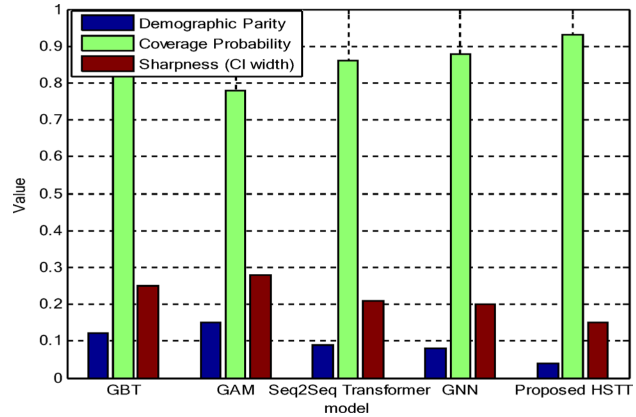

Fig. 6 Fairness and Calibration Analysis of Prediction Models

Table 7 Fairness & calibration

Model	Demographic Parity	Coverage Probability	Sharpness (CI width)
GBT	0.12	0.82	0.25
GAM	0.15	0.78	0.28
Seq2Seq Transformer	0.09	0.86	0.21
GNN	0.08	0.88	0.2
Proposed HSTT	0.04	0.93	0.15

training and 190s testing – was based on the fact that it is a multi-layer hierarchical learning model with a combination of temporal attention and dynamic graph learning. Although this requires longer runtime than that of the original, the high improvement in predictive accuracy, fairness, and calibration is worth the computational cost, particularly because training is not required (offline) and inference can be optimized to run in practice.

The predictive inputs in training were not based on demographic characteristics of the fairness and calibration analysis (Table 3). Rather, they were delimited at an aggregate spatial scale based on publicly available census-equivalent statistics that were identified with each spatial unit (e.g., precinct-level demographic composition). Variables like the race or age brackets were categorized in anonymized and rough terms so that no individual could be identified. The evaluation of fairness was performed post hoc, i.e., by comparing predicted risk distributions of crime in these aggregate groups of demographics based on demographic parity (D). Demographic validity checking was done based on consistency with official census summaries and the stability of group definitions over time. Such a distinction between model training and fairness auditing guarantees that fairness gains are based on model behavior and not explicit use of sensitive variables, which contributes to ethically and responsibly deploying models. Table 7 measures fairness and calibration measures. GBT (D = 0.12, coverage = 0.82, sharpness = 0.25) and GAM (D = 0.15, coverage = 0.78, sharpness = 0.28) were biased, and their calibration was less strong. The Seq2Seq Transformer achieved better fairness (D = 0.09, coverage = 0.86, sharpness = 0.21), and the GNN was better (D = 0.08, coverage = 0.88, sharpness = 0.20). The HSTT proposed gave the most significant results, where D = 0.04, the highest coverage (0.93), the lowest sharpness (0.15), and the results are fair and the confidence intervals are reliable. These results show that HSTT is superior in both the ethical orientation and forecast accuracy.

Figure 6 shows the fairness and calibration results of the models based on demographic parity, coverage probability, and sharpness (CI width). The lowest fairness of D = 0.15 and D = 0.78 coverage and the broadest intervals (sharpness = 0.28) of the GAM indicate the presence of both bias and uncertainty. GBT performed somewhat better, with D = 0.12, coverage = 0.82, and sharpness = 0.25; however, it was not very fair. Temporal modelling allowed the Seq2Seq Transformer to increase the balance by 0.09, covering 0.86, and sharpness by 0.21. The GNNs improved further, reaching D = 0.08, coverage 0.88, and sharpness 0.20, with the benefit of spatial neighborhood effects. The proposed HSTT exhibited the least disparity (D = 0.04), greatest coverage (0.93), and smallest intervals (0.15). This would not only bring fairness to the HSTT in terms of demographic boundaries but would also yield more realistic and accurate estimates of uncertainty, which would again bring fairness to the HSTT as an even more ethical predictive policing.

Figure 7 shows the ROC graph portrays the relative ability of the baseline models and the proposed HSTT to classify the NYPD data by drawing a graph showing the TPR versus the FPR. As indicated, GAM performs the worst, with an AUC of 0.76, meaning that it does not discriminate well because it utilizes smooth additive functions that are incapable of capturing the complex interactions of space and time crimes. GBT can achieve a performance comparable to that of AUC 0.80, with the advantage of learning in an ensemble but restricted to structured tabular patterns. The Seq2Seq Transformer achieved an AUC of 0.83, indicating

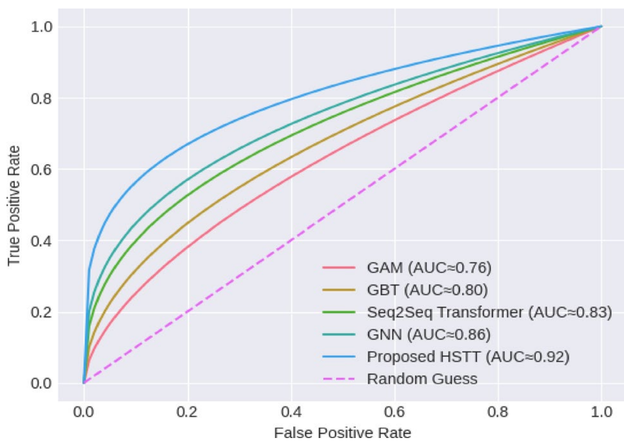

Fig. 7 ROC Comparison of the Baseline Models and the Proposed HSTT

better detection with long-range dependency modeling and temporal attention. The GNN also boosts discrimination stronger at 0.86 with an AUC of 0.86 and clearly identifies spatial contagion effects between adjacent areas. The proposed HSTT has the highest AUC value of 0.92 and a steady high TPR for all FPR levels. This advantage is due to its hierarchical integration of temporal attention and dynamic graph learning, which allows for the effective learning of both local and global crime propagation patterns.

Ablation Study: Impact of Temporal Attention and Graph Learning

To evaluate the role of individual architectural elements in the proposed Hierarchical Spatiotemporal Transformer (HSTT), an ablation study was performed with the NYPD dataset using the same preprocessing, training, and evaluation conditions. Based on the complete HSI model, major parts were chosen and removed or simplified, and all other hyperparameters were kept constant. Accuracy, F1-score, AUC, and RMSE were used as performance measures. The ablation configurations are less than.

- HSTT (Full Model): Temporal self-attention + dynamic graph attention + hierarchical fusion.
- HSTT - Graph Learning: Spatial dependencies were not removed, and levels of temporal attention were retained (no GAT).
- HSTT no Temporal Attention: Graph learning maintained; temporal encoder changed with fixed-window aggregation.
- Temporal-Only Model: Transformer-based temporal modelling without graph learning in the space.
- Graph-only model: A GNN that does not have time attention.

Table 8 Fairness & calibration

Model Variant	Accuracy	F1-score	AUC	RMSE
Graph-Only (GNN)	0.85	0.84	0.86	0.25
Temporal-Only (Seq2Seq)	0.82	0.81	0.83	0.29
HSTT w/o Temporal Attention	0.87	0.86	0.88	0.23
HSTT w/o Graph Learning	0.88	0.87	0.89	0.22
Proposed HSTT (Full)	0.91	0.90	0.92	0.18

The ablation results presented in Table 8 clearly indicate that both temporal attention and graph learning play a critical role in improving the performance of HSTT. The elimination of temporal attention also caused a significant decrease in AUC (0.92–0.88), which means that crime dynamics over long ranges and atypical time-dependent behavior were less effectively modeled. Similarly, the ability to perform spatial reasoning via graph learning is impaired by the absence of graph learning, which validates the necessity of accounting for neighborhood contagion and cross-district effects. Although single-component models (temporal only and graph only) perform better than the traditional baselines, both are not as good as the entire HSTT. The purest performance improvement is achieved by combining the temporal and spatial modules in a hierarchical manner, which proves the fundamental idea of the design of the proposed structure.

Conclusion

In conclusion, this paper proposed a hierarchical spatiotemporal urban crime prediction framework that provides a data-driven model combining temporal attention mechanisms with dynamic graph learning. The hybrid HSTT was always superior to either machine learning or standalone deep learning models, with the best balance of predictive accuracy, error minimization, fairness, and calibration. Generalized Additive Models and Gradient-Boosted Trees were useful and interpretable but too simple to capture complicated dynamics in cities. The HSTT was shown to be superior to Seq2Seq Transformers and Graph Neural Networks in terms of their respective capabilities to utilize contagion effects, cross-district dependencies, and irregular event regimes. Notably, HSTT was not biased in terms of demographics, and the estimates of uncertainty were quite decent, which matters because predictive policing carries significant ethical issues. The model is practical because it is trained offline and with optimized inference despite increased computation requirements. In general, in this paper, we have seen that responsible AI-driven crime prediction can be attained through the combination of accuracy, adaptability, interpretability, and ethical considerations. The framework can be extended to include multi-city information, real-time adaptive learning, and refinement of governance policies in the future to support equitable applications in the context of urban safety.

Future work will focus on improving real-time deployment feasibility by integrating streaming data pipelines, low-latency inference, and adaptive retraining for large-scale urban environments. Additionally, stronger ethical policy integration will be pursued through embedded fairness constraints, continuous bias auditing, and alignment with transparent governance frameworks to support responsible law enforcement decision-making.

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Data Availability No dataset was used or generated in the preparation of this manuscript.

Declarations

Conflict of interest The authors declare that there are no conflicts of interest related to this work.

Informed Consent There is no animal and human sample used.

Research Involving Human Participants and/or Animals This article does not contain any studies involving animals performed by any of the authors. This article does not contain any studies involving human participants performed by any of the authors.

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